

DCPO: DYNAMIC CLIPPING POLICY OPTIMIZATION

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ABSTRACT

Reinforcement Learning from Verifiable Rewards (RLVR) provides a scalable route for strengthening the reasoning capabilities of large language models by replacing learned preference models with rule-verifiable outcome rewards. Yet widely used RLVR optimizers such as GRPO can produce zero or ineffective gradients when fixed token-level probability-ratio bounds suppress informative low-probability tokens and stepwise reward standardization assigns identical rewards zero advantage. We introduce **Dynamic Clipping Policy Optimization (DCPO)**, a policy-optimization method that directly targets these two failure modes. DCPO combines a probability-aware dynamic clipping rule, which adapts token-level clipping bounds according to old-policy probabilities and thereby expands admissible exploration for rare tokens, with smooth advantage standardization, which incorporates cumulative reward statistics to improve the utilization of generated responses. Across four mathematical reasoning benchmarks and four Qwen model variants, DCPO establishes state-of-the-art or strongest average performance among the evaluated methods. On Qwen2.5-Math-7B, DCPO reaches 46.7 Avg@1 under greedy decoding and 38.8 Avg@32 under 32-sample decoding on AIME24, exceeding DAPO (36.7/31.6), GRPO (36.7/32.1), and GSPO (40.0/34.9). On Qwen2.5-14B, DCPO achieves 23.3/19.0 on AIME25, improving over GRPO (13.3/10.5), DAPO (20.0/15.3), and GSPO (16.7/9.9). DCPO also increases the average nonzero-advantage ratio by 28% over GRPO across four models, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude compared with GRPO and DAPO. These results show that DCPO converts generated data into effective RLVR updates more efficiently while preserving the token-level exploration needed for mathematical reasoning.

1 INTRODUCTION

The reasoning capabilities of large language models (LLMs) have become a central criterion for evaluating progress in model development (OpenAI, 2025; Guo et al., 2025; Comanici et al., 2025). Reinforcement Learning from Verifiable Rewards (RLVR) has emerged as a particularly effective optimization paradigm for this setting because it uses rule-based outcome rewards rather than subjective preference labels. This design is well aligned with domains in which correctness can be verified mechanically and multi-step deliberation is required, including mathematical problem solving and coding (DeepSeek-AI et al., 2024).

The GRPO family of algorithms (DeepSeek-AI et al., 2024; Liu et al., 2025; Yuan et al., 2025; Zheng et al., 2025) has become a widely used foundation for RLVR. Its empirical strength has been confirmed across several studies, but subsequent analyses, including DAPO (Yu et al., 2025), expose limitations that are especially consequential for reasoning-oriented training. GRPO can undergo entropy collapse, with advantage values decaying toward zero in later stages, and its fixed clipping rule can suppress rarely sampled but informative token-level updates. Together, these effects reduce the diversity of exploration precisely where RLVR depends on discovering and reinforcing sparse successful reasoning trajectories.

DAPO responds to these issues with Clip-Higher, which relaxes the upper clipping bound, and Dynamic Sampling, which filters prompts whose sampled responses receive uniform rewards so that each batch contains valid gradients. These interventions improve upon GRPO but leave two structural inefficiencies unresolved. First, dynamic sampling trades gradient availability for response-level sampling efficiency, which slows convergence because generated responses may be discarded and

regenerated. Second, a token-independent clipping interval remains mismatched to the probability structure of the policy: our analysis shows that the appropriate clipping range depends on each token’s old-policy probability, so a single uniform upper bound remains suboptimal even when it is made asymmetric. Related methods that avoid directly discarding zero-gradient responses do not remove these token-level and response-level constraints.

We introduce **Dynamic Clipping Policy Optimization (DCPO)**, a coordinated RLVR optimizer that addresses these limitations at the response, token, and loss-aggregation levels. First, DCPO uses a **smoothing technique that standardizes advantages across cumulative steps**, replacing purely step-local normalization with statistics that combine the current reward distribution with the distribution of responses generated for the same prompt over prior steps. This mechanism preserves learning signal when a current group receives identical rewards and thereby improves response-level exploration (see Section 2). Second, DCPO introduces a **dynamic clipping method that adaptively adjusts the clipping bounds based on the old token-specific probabilities**; rather than selecting another pair of fixed asymmetric thresholds, it widens the admissible update region for lower-prior-probability tokens, which are precisely the tokens for which exploration is most constrained under fixed clipping (see Section 2 and Appendix A.6). Finally, DCPO averages the loss only across tokens within each response, rather than across the entire batch, preserving the relative advantage ordering among responses to the same prompt while avoiding batch-level dilution.

We evaluate DCPO on a merged mathematical training set formed from Math-DAPO-17k (Yu et al., 2025) and levels 3–5 of the MATH dataset (Liu et al., 2025), using four mathematical reasoning benchmarks and four model backbones. Against GRPO, DAPO, and GSPO where available, DCPO delivers the strongest average performance across model scales, with especially clear gains on the challenging AIME benchmarks. Under 32-sample evaluation, DCPO-7B scores 38.8 on AIME24, compared with 32.1 for GRPO, 31.6 for DAPO, and 34.9 for GSPO; DCPO-14B reaches 19.0 on AIME25, compared with 10.5 for GRPO, 15.3 for DAPO, and 9.9 for GSPO. Beyond accuracy, DCPO raises the average nonzero-advantage ratio by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces token clipping by an order of magnitude compared with the principal baselines, demonstrating that its performance gains are accompanied by measurably more efficient use of generated data.

Appendix A.2 provides the preliminary definitions needed for this comparison, including the advantage and loss formulations used by GRPO, DAPO, and GSPO.

2 DYNAMIC-ADAPTIVE CLIPPING BOUNDS FOR RL(DAC)

Probability-ratio clipping is a standard stabilization device in reinforcement learning: it limits the magnitude of policy updates and thereby prevents abrupt parameter shifts during optimization. The mechanism is especially important for importance-sampling-based updates, where reweighting by the ratio $r(x)$ between the new probability $p(x)$ and the old probability $q(x)$ yields an unbiased estimate of $\mathbb{E}_{q(x)}[f(x) \frac{p(x)}{q(x)}]$ but may substantially inflate variance, as shown in Equation (1).

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p} [f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (1)$$

To control this variance inflation, prior work (e.g., Schulman et al. (2017)) constrains the probability ratio by a fixed hyperparameter ϵ , yielding the symmetric interval in Equation (2).

$$|r(x) - 1| \leq \epsilon \quad (2)$$

A symmetric fixed bound, however, imposes the same relative constraint on all tokens and therefore restricts exploration most severely in regions where the old policy assigns low probability mass. In such regions, the absolute amount of allowable probability movement is small, so potentially informative rare-token updates can be suppressed before they influence the policy. DAPO partially relaxes this constraint through asymmetric clipping, but the bound remains token independent. GSPO moves clipping to the sequence level; when the sequence-level importance ratio exceeds the asymmetric interval, the method discards entire responses, which can waste informative tokens while still retaining high-variance token-level behavior that destabilizes training.

From the perspective of controlling the variance–bias trade-off, the appropriate restriction should depend on the probability mass assigned by the old policy. We therefore impose a probability-dependent constraint, formalized in Equation (3).

$$|(r(x) - 1)p(x)| \leq \epsilon \quad (3)$$

Substituting $p(x) = r(x)q(x)$ into Equation (3) gives a closed-form admissible range for $r(x)$ once the square-root term is made non-negative by taking its maximum with zero. The resulting **dynamic-adaptive clipping bounds** depend explicitly on $q(x)$ and on the hyperparameters ϵ_{low} and ϵ_{high} , as specified in Equation (4).

$$0.5 + \frac{1}{2} \sqrt{\max \left(1 - \frac{4\epsilon_{low}}{q(x)}, 0 \right)} \leq r(x) \leq 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (4)$$

This construction enlarges the allowable update interval precisely in low-probability regions, where fixed clipping is most restrictive, while retaining tighter control for high-confidence tokens. Following the dual-clipping motivation of (Ye et al., 2020), we cap the maximum ratio for both positive and negative advantages at 10 to prevent excessively large adaptive bounds from producing unstable updates. Appendix A.5 and Appendix A.6 give the full derivation and analyze the resulting bound geometry from the importance-sampling perspective.

3 SMOOTH ADVANTAGE STANDARDIZATION(SAS)

GRPO, DAPO, and GSPO compute the advantage $\hat{A}_{new,j}^i$ of the j -th response by standardizing the rewards $R_{j=1,\dots,G}^i$ sampled for the same prompt at the i -th step, as in Equation (11) of Appendix A.3. This step-local standardization creates two failure modes. If sampling randomness produces identical rewards for all responses in a group, every standardized advantage becomes zero, and the prompt contributes no parameter update even when the sampled reasoning trajectories differ. Conversely, high-entropy sampling can produce highly skewed label counts, causing standardized advantages to fluctuate sharply across steps and even change sign, which destabilizes optimization.

Recent evidence (Yue et al., 2025) further suggests that RLVR primarily adjusts the likelihood of correct samples already present in the base model’s sampling distribution, rather than fundamentally replacing that distribution. Under this view, responses generated for the same prompt across training steps provide a coherent reward history. DCPO therefore uses *cumulative* standardization, defined in Equation (5), to incorporate the reward distribution observed for that prompt over training.

$$\hat{A}_{total,j}^i = \frac{(R_j^i - \mu_{total}^i)}{\sigma_{total}^i} \quad (5)$$

where the mean and standard deviation are computed over all responses generated so far for the same prompt.

To moderate the instability of both step-specific standardization ($\hat{A}_{new,j}^i$) and cumulative standardization ($\hat{A}_{total,j}^i$), we define two smoothed quantities, $\hat{S}A_{new,j}^i$ and $\hat{S}A_{total,j}^i$. Each is a step-dependent weighted average of the current and cumulative standardized advantages, with the weights changing as a function of step i in Equation (6).

$$\hat{S}A_{new,j}^i = \frac{i-1}{i} \hat{A}_{new,j}^i + \frac{1}{i} \hat{A}_{total,j}^i, \quad \hat{S}A_{total,j}^i = \frac{1}{i} \hat{A}_{new,j}^i + \frac{i-1}{i} \hat{A}_{total,j}^i \quad (6)$$

The quantity $\hat{S}A_{new,j}^i$ increasingly emphasizes the current distribution as training proceeds, allowing the advantage estimate to track recent policy outputs. By contrast, $\hat{S}A_{total,j}^i$ emphasizes the cumulative distribution and treats the current step as a local deviation within the broader reward history. DCPO then defines the final advantage $\hat{A}_j^i$ as the smoothed value with the smaller absolute

magnitude, as in Equation (7), thereby reducing the influence of fluctuations introduced by either purely current-step or purely cumulative standardization.

$$\hat{A}_j^i = \begin{cases} \hat{S}A_{new,j}^i, & \text{when } |\hat{S}A_{new,j}^i| < |\hat{S}A_{total,j}^i| \\ \hat{S}A_{total,j}^i, & \text{otherwise} \end{cases} \quad (7)$$

After a prompt has entered optimization, responses generated for that prompt in subsequent steps remain eligible for model updates even when the current stepwise advantage is zero. Thus, when all rewards within a step are identical, DCPO still supplies an update through the cumulative component $\frac{1}{i}\hat{A}_{total,j}^i$, preserving otherwise lost learning signal and improving response-level data efficiency.

3.1 OTM LOSS

The **Token-Level Mean loss (TLM)** used by DAPO in Equation (13) equalizes token contributions across the batch, but in doing so it can distort the intended ordering among responses of different lengths. Consider two responses to the same prompt: response A has advantage 1 and length 500, whereas response B has advantage 0.5 and length 1500. Under TLM, A receives weight $\frac{500}{500+1500} = 0.25$ and B receives weight $\frac{1500}{500+1500} = 0.75$, so their total loss contributions become $1 \times 0.25 = 0.25$ and $0.5 \times 0.75 = 0.375$, respectively. The lower-advantage response therefore exerts the larger influence solely because it is longer.

The **Sequence-Level Mean loss (SLM)** used by GRPO in Equation (12) introduces the complementary problem: averaging across the G responses attenuates the magnitude of the already standardized advantages, effectively reducing their scale by G . This attenuation becomes more consequential as the batch size grows and can weaken the relative structure encoded by advantage standardization. DCPO therefore treats standardized advantages as already defining the relative ordering among responses, averages only over the tokens within each response through $|o_i|$, and does not average across the responses in the batch. This **Only Token Mean loss (OTM)**, given in Equation (8), preserves response-level advantage relationships while distributing each response’s contribution uniformly over its tokens.

$$\mathcal{T}_{\text{DCPO}}(\theta) = \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (8)$$

As in DAPO, DCPO does not include an explicit KL-divergence constraint in the loss. Its lower and upper clipping limits are supplied by the dynamic-adaptive bounds introduced in Section 2.

4 EXPERIMENTAL SETUP

Models and Datasets We evaluate DCPO on four Qwen backbones spanning instruct, common-base, and math-base settings: Qwen2.5-Math-1.5B-Instruct, Qwen2.5-3B (common base), Qwen2.5-Math-7B (math base), and Qwen2.5-14B (common base) (Yang et al., 2024a;b). Training uses a merged corpus of approximately 25k mathematical problems, constructed by combining DAPO-Math-17K (Yu et al., 2025) with levels 3–5 of the MATH subdataset (Hendrycks et al., 2021b; Liu et al., 2025). For consistency across training and evaluation, all problems are formatted with the Qwen-Math template (see Appendix A.8).

Evaluation Evaluation covers four established mathematical reasoning benchmarks: MATH500 (Hendrycks et al., 2021a), AMC23 (Ouyang et al., 2022), AIME24 (Li et al., 2024), and AIME25 (MAA). Because MATH500 contains 500 problems, we report Avg@1 for that benchmark. For the smaller AMC23 (40 problems), AIME24 (30 problems), and AIME25 (30 problems) benchmarks, we report both Avg@1 and Avg@32 so that deterministic accuracy and sampling-based robustness are both visible.

- **Avg@1:** Accuracy under greedy decoding, measuring the model’s deterministic prediction for each problem.
- **Avg@32:** Average accuracy over 32 sampled responses per problem with temperature 1.0 and top_p 1.0, measuring the robustness and stability of the learned policy distribution under stochastic decoding.

Baseline Settings GRPO and DAPO serve as the primary baselines, and GSPO is included as an additional baseline for Qwen2.5-Math-7B and Qwen2.5-14B. Appendix A.9 specifies the remaining experimental settings, including clipping thresholds ϵ_* , codebase details, and shared training hyperparameters.

5 RESULTS AND ANALYSIS

5.1 MAIN RESULTS

Table 1: Benchmark performance for DCPO and baselines across model scales. Avg@1 denotes greedy decoding accuracy, Avg@32 denotes average accuracy over 32 sampled responses, boldface marks the best value in each comparison block, and Δ reports DCPO’s difference from the corresponding baseline.

Model	MATH500 (Avg@1)	AMC23 (Avg@1/32)	AIME24 (Avg@1/32)	AIME25 (Avg@1/32)	Average (Avg@1/32)
Qwen2.5-Math-1.5B-Instruct					
base	73.6	57.5/49.4	10.0/10.0	3.3/6.1	36.1/21.8
GRPO	77.2	70.0/68.4	16.7/14.0	20.0/13.5	46.0/32.0
DAPO	76.0	80.0/70.6	20.0/13.5	14.4/12.5	46.5/32.4
DCPO	77.2	75.0/ 70.8	20.0/15.6	16.7/12.1	47.2/32.8
Δ_{GRPO}	+0.0	+5.0/+2.4	+3.3/+1.6	-3.3/-1.4	+1.2/+0.8
Δ_{DAPO}	+1.2	-5.0/ +0.2	+0.0/+2.1	+2.3/-0.4	+0.7/+0.4
Qwen2.5-3B[Base]					
base	46.4	27.5/7.8	3.3/0.1	3.3/0.7	20.1/2.9
GRPO	69.2	62.5/51.6	10.0/7.5	6.7/4.5	36.3/21.0
DAPO	72.4	57.5/54.0	10.0/8.3	6.7/3.9	36.6/23.1
DCPO	71.2	62.5/55.8	3.3/7.5	10.0/4.7	37.6/22.7
Δ_{GRPO}	+2.0	+0.0/+4.2	-6.7/ +0.0	+3.3/+0.2	+1.3/+1.7
Δ_{DAPO}	-1.2	+5.0/+1.8	-6.7/-0.8	+3.3/+0.8	+1.0/-0.4
Qwen2.5-Math-7B[Base]					
base	50.4	40.0/19.5	13.3/6.0	3.3/1.5	28.4/9.3
GRPO	81.6	77.5/75.9	36.7/32.1	16.7/16.7	53.1/41.6
DAPO	83.0	72.5/ 80.7	36.7/31.6	23.3/14.9	53.9/42.4
GSPO	84.0	80.0/78.8	40.0/34.9	16.7/16.2	55.2/43.3
DCPO	82.5	82.6/79.8	46.7/38.8	16.7/ 17.2	57.1/45.2
Δ_{GRPO}	+0.9	+5.1/+4.9	+10.0/+6.7	+0.0/+0.5	+4.0/+3.6
Δ_{DAPO}	-0.5	+10.1/-0.9	+10.0/+7.2	-6.6/ +2.3	+3.3/+2.8
Δ_{GSPO}	-1.5	+2.6/+1.0	+6.7/+3.9	+0.0/+1.0	+1.9/+1.9
Qwen2.5-14B[Base]					
base	60.8	47.5/16.4	3.3/1.3	3.3/1.1	28.7/6.3
GRPO	81.2	75.0/65.6	13.3/17.6	13.3/10.5	45.7/31.3
DAPO	83.4	87.5/85.1	16.7/16.4	20.0/15.3	51.9/38.9
GSPO	78.6	77.5/75.0	23.3/16.0	16.7/9.9	49.0/33.5
DCPO	84.6	85.0/79.9	20.0/ 18.2	23.3/19.0	53.2/39.0
Δ_{GRPO}	+3.4	+10.0/+14.3	+6.7/+0.6	+10.0/+8.5	+6.5/+7.7
Δ_{DAPO}	+1.2	-2.5/-5.2	+3.3/+1.8	+3.3/+3.7	+1.3/+0.1
Δ_{GSPO}	+6.0	+7.5/4.9	-3.3/ +2.2	+6.6/+10.1	+4.2/+5.5

Table 1 reports the central comparison among GRPO, DAPO, GSPO, and DCPO on MATH500, AMC23, AIME24, and AIME25. The table separates deterministic performance under greedy decoding (Avg@1) from sampling-based performance (Avg@32), making clear whether each optimizer improves only the most likely output or also strengthens the broader policy distribution. Across the four evaluated model scales, DCPO achieves the strongest average results or remains competitive with the best baseline, establishing it as an effective RLVR policy-optimization method for mathematical reasoning.

Overall Superiority of DCPO On Qwen2.5-Math-1.5B-Instruct, DCPO obtains the highest average scores, 47.2/32.8, improving over GRPO by +1.2/+0.8 and over DAPO by +0.7/+0.4. On Qwen2.5-3B, DCPO reaches 37.6/22.7, exceeding GRPO by +1.3/+1.7 and remaining competitive with DAPO, with a +1.0 Avg@1 gain and only a -0.4 Avg@32 difference. The gains become more pronounced at larger scales: on Qwen2.5-Math-7B, DCPO achieves 57.1/45.2, surpassing GRPO by +4.0/+3.6, DAPO by +3.3/+2.8, and GSPO by +1.9/+1.9; on Qwen2.5-14B, it reaches 53.2/39.0, improving over GRPO by +6.5/+7.7, DAPO by +1.3/+0.1, and GSPO by +4.2/+5.5. These comparisons show that DCPO’s advantage is not confined to a single backbone or decoding mode, but appears across model capacities under both greedy and stochastic evaluation.

Robustness on AIME under Sampling (Avg@32) The clearest stress-test results occur on AIME24 and AIME25, the most difficult competition-style benchmarks in the evaluation. Avg@32 measures whether the trained policy can repeatedly sample correct solutions rather than merely place a single correct solution at the greedy mode. Under this metric, DCPO is especially strong at the 7B and 14B scales: on AIME24-Avg@32, DCPO-7B scores 38.8, compared with 32.1 for GRPO, 31.6 for DAPO, and 34.9 for GSPO; on AIME25-Avg@32, DCPO-14B reaches 19.0, compared with 10.5 for GRPO, 15.3 for DAPO, and 9.9 for GSPO. These results indicate that DCPO improves not only point-estimate accuracy but also the reliability with which correct reasoning trajectories remain available under high-entropy sampling.

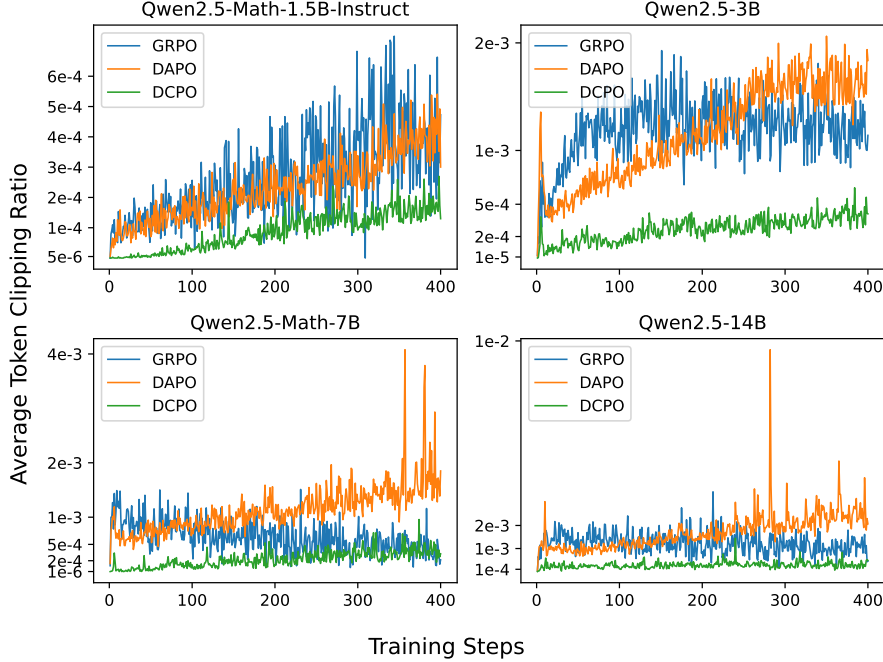


Figure 1: Token Clipping Ratio (TCR) across model scales and optimization methods; lower values indicate that fewer generated tokens are excluded from back-propagation by clipping.

5.2 THE TOKEN CLIPPING RATIO (TCR)

We define the **Token Clipping Ratio (TCR)** as the proportion of generated tokens that are excluded from policy updates during back-propagation because their probability ratios are clipped. Figure 1 shows that this quantity sharply distinguishes the optimization behavior of the evaluated RLVR methods.

$$\text{TCR} = \text{Average} \left(\sum_{m=1}^N \frac{\text{Number of clipped tokens in } \textit{micro}_m}{\text{Total number of tokens in } \textit{micro}_m} \right) \quad (9)$$

where $\textit{micro}_m$ denotes the m -th microbatch and N is the number of microbatches per training step.

GRPO exhibits model-dependent TCR dynamics: the ratio increases over training for Qwen2.5-Math-1.5B and Qwen2.5-3B, but gradually decreases for Qwen2.5-Math-7B and Qwen2.5-14B. DAPO,

by contrast, shows an upward *TCR* trajectory on all four models, indicating that an increasing share of generated tokens is removed from effective policy updates as training proceeds. Both GRPO and DAPO also display substantial fluctuations and occasional spikes, patterns that can destabilize optimization and are consistent with the entropy-collapse behavior discussed earlier. GSPO has an average *TCR* above 11% on Qwen2.5-Math-7B and above 15% on Qwen2.5-14B, substantially higher than the token-level clipping methods and indicative of greater loss of informative generated content.

DCPO maintains a markedly more stable clipping profile. Across model sizes and training stages, its *TCR* remains relatively constant and approximately an order of magnitude lower than those of GRPO and DAPO. This low clipping ratio means that more generated tokens remain active in policy updates, so complete response sequences contribute more fully to learning while the adaptive bounds still control outlier probability-ratio variation. The *TCR* evidence therefore supports the central design claim of DCPO: probability-aware clipping improves sample efficiency by preserving rare-token learning signal without abandoning update stability.

Qwen2.5-Math-7B illustrates why this distinction matters. The token distribution rapidly concentrates on high-confidence tokens: after approximately 60 steps, about 95% of generated tokens satisfy $q(x) > 0.9$; after approximately 100 steps, this fraction rises to about 97% and continues to increase later in training. As shown in Figure 4d, once the old probability satisfies $(q(x) \geq \frac{1}{1+\epsilon_{grpo}} \approx 0.83)$, the new probability $p(x)$ remains eligible for model updates in the interval $[\frac{1}{1+\epsilon}, 1]$ under any of the clipping methods. Thus, most tokens occupy a region that is rarely clipped, whereas the tokens most affected by clipping are the low-probability tail. The later-stage increases in *TCR* in Figure 1 therefore correspond to a growing risk that rare tokens are excluded from learning, directly limiting exploration diversity.

This observation aligns with prior evidence that **high-entropy tokens** are important drivers of emergent reasoning capabilities (Wang et al., 2025). Because these tokens are also low-probability under the current policy, they carry disproportionate exploratory information. A broader admissible update range for such tokens therefore enlarges the effective exploration space available to RLVR and supports the acquisition of more diverse reasoning trajectories.

Together, the empirical *TCR* patterns and the entropy-based interpretation justify the **dynamic-adaptive clipping** scheme. DCPO relaxes clipping for tokens with $q(x) < \frac{1}{1+\epsilon}$ while preserving the conventional safe region for high-confidence tokens. By selectively expanding the update interval for the high-entropy tail, DCPO restores access to rare but informative tokens and thereby improves the conditions under which reasoning ability can be reinforced.

5.3 THE RESPONSE UTILIZATION RATIO (*RUR*)

In RLVR, a generated response contributes to optimization only when it induces a nonzero gradient. We measure this response-level efficiency with the **Response Utilization Ratio (*RUR*)**, defined as the percentage of generated responses with nonzero advantage that participate in the policy update.

$$RUR = \frac{\text{Number of responses with non-zero advantage}}{\text{Total number of generated responses}} \times 100\% \quad (10)$$

Table 2 shows that GRPO uses generated responses inefficiently: its overall average *RUR* is only 44.6% across the four models, so more than half of sampled responses fail to contribute effective gradients. DCPO raises the average *RUR* to approximately 70% after the first epoch, an absolute gain of 28% and a relative improvement of 64% over GRPO. This result directly supports the role of smooth advantage standardization in retaining response-level learning signal that step-local standardization would otherwise erase.

Table 2: Average *RUR* over 400 training steps; higher values indicate more effective response-level utilization.

model	GRPO	GSPO	DCPO
Qwen2.5-Math-1.5B-Instruct	45.6%	-	67.1%
Qwen2.5-3B	48.3%	-	74.3%
Qwen2.5-Math-7B	37.4%	43.5%	73.2%
Qwen2.5-14B	43.9%	47.6%	72.4%
Average	43.8%	45.6%	71.8%

Figure 2 shows that GRPO’s *RUR* decreases substantially over training across model sizes and model types, falling from above 90% to below 50%. On Qwen2.5-Math-7B, for example, *RUR* falls to roughly 30% after approximately 200 training steps. This pattern indicates that current-step advantage standardization increasingly produces zero-gradient responses, making later-stage exploration inefficient. DCPO, by contrast, maintains a high *RUR*: after the first epoch, the ratio stabilizes around 70% and continues to improve as training proceeds.

The early *RUR* values above 90% and the later stabilization around 70% have different interpretations. The initial high ratio largely reflects early model outputs that fail to satisfy the required format, receive negative rewards, and therefore inflate the count of nonzero advantages. Once models learn the fixed answer format, response utilization becomes a more direct measure of reasoning-relevant exploration.

The remaining gap between 70% and 100% is attributable to two forms of persistently zero advantage. (i) Nearly half of the mathematical problems are sufficiently easy that the model solves them consistently, so excluding them from further policy updates is reasonable because the corresponding capability requires little additional refinement. (ii) Other instances are consistently incorrect, either because they exceed the current model’s capability or because some DAPO-Math-17K labels were converted into mislabeled integers. In these cases, zero-advantage exclusion can function as a useful filter, reducing the risk of updating on noisy or mislabeled supervision.

DAPO is not directly comparable under *RUR*, because it discards generated responses with identical labels before the update. We therefore evaluate DAPO’s training efficiency by measuring the number of generated responses and the training time required to complete the same number of parameter-update steps. To compensate for GRPO-like low sampling efficiency, DAPO uses a generation batch size between one and four times the training batch size, adjusted dynamically during training. Across the four models, DAPO requires 3–5 times more generated responses than DCPO to obtain the same number of parameter updates, at least doubling GPU hours without producing better performance. Its utilization also deteriorates with scale: for Qwen2.5-14B, the proportion of DAPO responses used falls below 30% after about 300 steps.

These findings show that DCPO improves over GRPO and DAPO at the response-utilization level as well as at the accuracy level. By retaining more generated responses as effective updates, DCPO gives the policy broader opportunity to explore diverse reasoning trajectories while using training compute more efficiently.

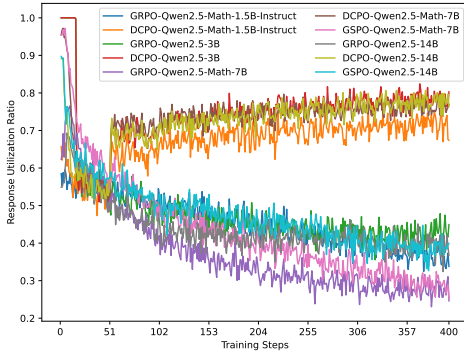


Figure 2: Training trajectory of *RUR* for GRPO and DCPO, showing the sustained response utilization achieved by smooth advantage standardization.

5.4 ABLATION STUDIES

We conduct an ablation study on Qwen2.5-Math-7B to isolate the contribution of each DCPO component under **Avg@32**, the metric most sensitive to robustness of the learned sampling distribution. For comparability, each ablation changes one component of the GRPO framework while holding the remaining settings fixed, and the KL-divergence term is removed to align the ablated variants with DAPO, GSPO, and full DCPO. Figure 3 reports the learning curves from 20 to 400 training steps.

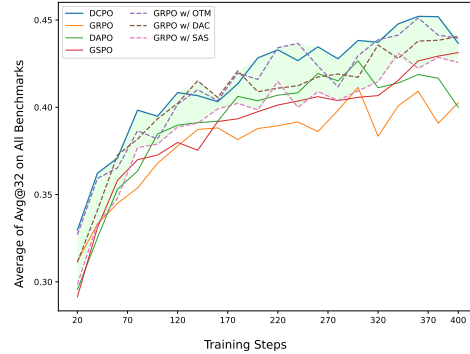


Figure 3: Ablation on Qwen2.5-Math-7B using average Avg@32, isolating the contributions of OTM, SAS, DAC, and their full DCPO combination.

The ablation curves show that each of the three components improves inference performance relative to the modified GRPO baseline, and the individual variants reach or exceed the performance of DAPO and GSPO in this setting. More importantly, the full combination yields the most stable and highest trajectory. Whereas the baselines plateau in later training, DCPO continues to improve, indicating that OTM, SAS, and DAC address complementary bottlenecks: response-level loss aggregation, advantage utilization, and token-level clipping. The component-level results therefore substantiate the design of DCPO as an integrated optimizer rather than a single isolated modification.

(i) **Only-Token-Mean loss (OTM)**: Aggregating the loss within each response rather than across the full batch improves the modified GRPO baseline, consistent with the claim that response-level advantage relationships should not be diluted by batch-level averaging. (ii) **Smoothed Advantage Standardization (SAS)**: Replacing purely stepwise advantages with cumulative smoothed advantages substantially improves GRPO and approaches DAPO, showing that reward statistics accumulated across training steps stabilize optimization and increase usable response signal. (iii) **Dynamic Adaptive Clipping (DAC)**: Replacing GRPO’s fixed clipping boundary with probability-adaptive bounds yields the strongest single-component gain; DAC alone surpasses GRPO, DAPO, and GSPO in this ablation and is more stable than OTM alone. (iv) **Combination of OTM + SAS + DAC (full DCPO)**: Combining the three mechanisms produces the best and most stable overall result, confirming that their benefits are complementary and jointly improve data-efficient, stable RLVR training for large language models.

Overall, the ablation study verifies that every DCPO component contributes positively and that their integration yields cumulative gains beyond any single modification. This evidence connects the method’s technical design to its empirical advantage in data efficiency, stability, and sampling-based mathematical reasoning performance.

6 CONCLUSION

This work identifies and addresses two central inefficiencies in existing RLVR policy optimization: fixed clipping restricts token-level exploration in low-probability regions, and current-step reward standardization wastes generated responses when rewards within a step are identical or unstable. **Dynamic Clipping Policy Optimization (DCPO)** resolves these issues through an integrated design: dynamically adaptive token-level clipping adjusts bounds according to old-policy probabilities, giving rare tokens a larger admissible update region, while **Cumulative-Smooth Advantage Standardization** incorporates reward statistics across optimization steps to preserve response-level learning signal under high-entropy sampling. Experiments on four mathematical reasoning benchmarks and four model backbones demonstrate that these mechanisms translate into stronger accuracy, higher response utilization, faster training, and substantially lower token clipping. In particular, DCPO-7B achieves 38.8 Avg@32 on AIME24, outperforming GRPO (32.1) and DAPO (31.6), while DCPO overall increases response utilization by 28% over GRPO, doubles training efficiency relative to DAPO, and reduces the token clipping ratio by an order of magnitude. These findings establish DCPO as a more data-efficient and exploration-preserving RLVR optimizer within the evaluated mathematical reasoning setting. Future work will investigate whether the same mechanisms extend to other verifiable-reward domains, including code generation and semantic reasoning.

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A APPENDIX

A.1 MAIN SYMBOL DEFINITIONS

G	Number of responses generated per prompt at each step.
θ	Parameters of the actor model.
$\pi_{\theta}(o_i q)$	Probability of response o_i given prompt q under parameters θ .
$\pi_{\theta_{old}}(o_i q)$	Probability of response o_i given prompt q under previous parameters θ_{old} .
π_{ref}	Reference policy for KL divergence computation.
β	Coefficient for KL divergence term (controls policy update vs. constraint trade-off).
$r_{i,t}(\theta)$	Probability ratio $\frac{\pi_{\theta}(o_i^t q, o_i^{1:t-1})}{\pi_{\theta_{old}}(o_i^t q, o_i^{1:t-1})}$ for token t in response o_i .
$\hat{A}_{i,t}$	Standardized advantage estimate for token t in response o_i .
$ o_i $	The length of the response in tokens.
$\mathbb{D}_{KL}(\pi_{\theta} \pi_{ref})$	KL divergence between current and reference policies.
ε	Probability ratio clipping threshold.
ε_{low} and ε_{high}	Lower/upper thresholds for probability ratio clipping.
M	Maximum allowed response length.
$is_equivalent(a, o_i)$	Binary function indicating whether response o_i matches answer a .
μ_{new}^i	Mean reward across G responses generated at step i (0 if $i = 0$).
σ_{new}^i	Reward variance across G responses at step i (0 if $i = 0$).
μ_{old}^i	Mean reward across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
σ_{old}^i	Reward variance across $G \cdot (i - 1)$ responses from steps 1 to $i - 1$ (0 if $i \leq 1$).
μ_{total}^i	Mean reward across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).
σ_{total}^i	Reward variance across $G \cdot i$ responses from steps 1 to i (0 if $i = 0$).

A.2 PRELIMINARY

A.3 ADVANTAGE CALCULATION

In GRPO and DAPO, the token-level advantage $\hat{A}_{j,t}^i$ for token t in response j is obtained by standardizing the response reward R_j^i with respect to the mean μ^i and standard deviation σ^i of the G responses generated for the same prompt at step i , as shown in Equation (11).

$$\hat{A}_{j,t}^i = \frac{(R_j^i - \mu^i)}{\sigma^i} \quad (11)$$

When all responses for the same prompt receive identical rewards, the standardized advantages are zero and the corresponding responses provide no model update, creating response-level waste.

A.3.1 GRPO

Shao et al. (2024) proposed **Group Relative Policy Optimization (GRPO)**, which samples G responses for each query, assigns rewards R with a rule-based reward function, and estimates token-level advantages according to Equation (11). Model parameters are then updated with the **Sequence Level Mean loss (SLM)** in Equation (12):

$$\mathcal{T}_{\text{GRPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_{i,t} \right) - \beta \mathbb{D}_{\text{KL}}(\pi_{\theta} \parallel \pi_{\text{ref}}) \quad (12)$$

The first term averages token-level contributions within each response and then across the G responses in the batch. The $\mathbb{D}_{\text{KL}}$ penalty limits deviation from a reference policy, and the fixed clipping threshold ε constrains large token-probability-ratio updates. This averaging strategy is widely adopted, but it also dilutes the magnitude and relative importance of individual responses.

A.3.2 DAPO

Yu et al. (2025) proposed **Dynamic sAmpling Policy Optimization (DAPO)** after observing that longer GRPO responses can exert disproportionate influence on gradient updates. DAPO normalizes token contributions across the full batch, producing the **Token Level Mean loss (TLM)** in Equation (13).

$$\mathcal{T}_{\text{DAPO}}(\theta) = \frac{1}{\sum_{i=1}^G |o_i|} \sum_{i=1}^G \sum_{t=1}^{|o_i|} \min \left(r_{i,t}(\theta) \hat{A}_{i,t}, \text{clip}(r_{i,t}(\theta), 1 - \varepsilon_{\text{low}}, 1 + \varepsilon_{\text{high}}) \hat{A}_{i,t} \right) \quad (13)$$

subject to $0 < |\{o_i \mid \text{is_equivalent}(a, o_i)\}| < G$. Prompts whose responses all receive identical rewards are discarded and resampled to maintain the batch size, which improves gradient availability but creates substantial data inefficiency.

A.3.3 GSPO

Zheng et al. (2025) proposed **Group Sequence Policy Optimization (GSPO)**, motivated by the concern that token-level clipping can introduce high-variance noise into gradient estimates. Because such noise can accumulate over long sequences and be amplified by clipping, GSPO replaces token-wise clipping with sequence-wise clipping to obtain a more stable training signal.

$$\mathcal{T}_{\text{GSPO}}(\theta) = \frac{1}{G} \sum_{i=1}^G \min \left(s_i(\theta) \hat{A}_i, \text{clip}(s_i(\theta), 1 - \varepsilon, 1 + \varepsilon) \hat{A}_i \right) \quad (14)$$

$$\text{where } s_i(\theta) = \exp \left(\frac{1}{|o_i|} \sum_{t=1}^{|o_i|} \log \left(\frac{\pi_{\theta}(o_{j,t}|q)}{\pi_{\theta_{\text{old}}}(o_{j,t}|q)} \right) \right)$$

In the reported settings, GSPO discards more than 10% of responses with nonzero advantages, increasing data waste and reducing training efficiency. GRPO, by contrast, does not remove these sequence-level high-variance responses and can retain many token-level high-entropy updates, which contributes to training instability.

A.4 REWARD CALCULATION

To provide a verifiable and interpretable supervision signal, we use a rule-based reward model that evaluates both answer correctness and format compliance, as defined in Equation (15). A response receives +1 when both format and answer are correct, 0 when the format is correct but the answer is wrong, and -1 when the required format is violated.

$$R_j^i = \begin{cases} 1, & \text{correct format and correct answer} \\ 0, & \text{correct format but incorrect answer} \\ -1, & \text{incorrect format} \end{cases} \quad (15)$$

where j indexes the response group for a given problem at training step i , and the reward is determined solely by response format and answer correctness.

A.5 HYPOTHETICAL REASONING PROCESS FOR DYNAMIC-ADAPTIVE CLIPPING BOUNDS

Probability-ratio clipping stabilizes reinforcement-learning updates by preventing large, destabilizing parameter changes. It is especially important when importance sampling estimates expectations under a new policy from samples generated by an old policy.

The expectation of a function $f(x)$ under the new probability $p(x)$ can be rewritten as an expectation under the old probability $q(x)$ using the standard importance weight, as shown in equation 16:

$$\int f(x)p(x) dx = \int f(x) \frac{p(x)}{q(x)} q(x) dx = \mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \quad (16)$$

Although this estimator is unbiased, it can substantially inflate variance, which is the central difficulty addressed by clipping. The true variance of $f(x)$ under the new policy is given by Equation (17), whereas the variance of the importance-sampled estimator under the old policy is given by Equation (18):

$$\text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p}[f(x)^2] - (\mathbb{E}_{x \sim p}[f(x)])^2 \quad (17)$$

$$\begin{aligned} \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] &= \mathbb{E}_{x \sim q} \left[\left(f(x) \frac{p(x)}{q(x)} \right)^2 \right] - \left(\mathbb{E}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \right)^2 \\ &= \mathbb{E}_{x \sim p} \left[f(x)^2 \frac{p(x)}{q(x)} \right] - (\mathbb{E}_{x \sim p}[f(x)])^2 \end{aligned} \quad (18)$$

The variance gap in Equation (19) makes the source of this inflation explicit:

$$\text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] - \text{Var}_{x \sim p}[f(x)] = \mathbb{E}_{x \sim p} \left[f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) \right] = \int f(x)^2 \left(\frac{p(x)}{q(x)} - 1 \right) p(x) dx \quad (19)$$

To reduce this variance gap, PPO Schulman et al. (2017) and RLVR methods (DeepSeek-AI et al., 2024; Liu et al., 2025) commonly impose a fixed symmetric bound on the probability ratio $r(x) = \frac{p(x)}{q(x)}$, as defined in Equation (20):

$$|r(x) - 1| \leq \epsilon \quad (20)$$

A fixed bound nevertheless restricts exploration most strongly where the old policy assigns low probability mass. In these regions, the absolute probability movement permitted by a fixed relative interval is small, so meaningful updates to rare tokens can be suppressed. DAPO’s asymmetric clipping relaxes one side of the interval, but the constraint remains non-adaptive with respect to token probability.

Recent studies such as Yue et al. (2025) suggest that RLVR primarily reweights reasoning paths already present in the base model’s sampling distribution. The small fraction of clipped parameters observed during off-policy batch updates is consistent with this view, indicating limited parameter shift. We therefore replace the conventional fixed clipping condition in Equation (20) with the probability-aware restriction in Equation (21), which constrains $r(x)$ through a dynamic-adaptive mechanism that incorporates the probability itself.

$$\left| \left(\frac{p(x)}{q(x)} - 1 \right) p(x) \right| \leq \epsilon \quad (21)$$

Substituting $p(x) = r(x)q(x)$ into the preceding inequality yields Equation (22).

$$-\epsilon \leq (r(x) - 1)r(x) * q(x) \leq \epsilon \quad (22)$$

For the resulting clipping ratio to be mathematically meaningful, the quantities must satisfy the probability and ratio constraints in Equation (23): $p(x)$ and $q(x)$ lie in $[0, 1]$, the ratio $r(x)$ is non-negative, and the clipping hyperparameter ϵ is non-negative.

$$\begin{cases} 0 \leq p(x) \leq 1 \\ 0 \leq q(x) \leq 1 \\ 0 \leq r(x) \\ 0 \leq \epsilon \end{cases} \quad (23)$$

Solving the inequality in Equation (22) under these constraints gives the candidate lower and upper boundary functions in Equation (24).

$$\begin{cases} low_{ge}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ high_{le}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon}{q(x)}} \\ low_{le}(q(x), \epsilon) = 0.5 - \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \\ high_{ge}(q(x), \epsilon) = 0.5 + \frac{1}{2}\sqrt{1 - \frac{4\epsilon}{q(x)}}, & \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{cases} \quad (24)$$

Combining Equation (22) with the boundary functions in Equation (24) yields the following admissible conditions for the probability ratio $r(x)$:

$$\begin{cases} low_{ge}(q(x), \epsilon) \leq 0 \leq r(x) \leq high_{le}(q(x), \epsilon), & \text{when } \text{Var}_{x \sim q} \left[f(x) \frac{p(x)}{q(x)} \right] \geq \text{Var}_{x \sim p} [f(x)] \\ r(x) \leq low_{le}(q(x), \epsilon) \cup high_{ge}(q(x), \epsilon) \leq r(x), & \text{otherwise} \end{cases} \quad (25)$$

To construct a practical clipping interval, we use the overlapping region centered around the ratio value 1. This choice minimizes the discrepancy between the two cases and preserves compatibility with GRPO's conventional fixed bound, leading to the validity condition below:

$$\begin{aligned} high_{ge}(q(x), \epsilon) \leq r(x) \leq high_{le}(q(x), \epsilon) \\ \text{S.t. } q(x)^2 - 4q(x) * \epsilon \geq 0 \end{aligned} \quad (26)$$

For low_{le} and $high_{ge}$, non-negativity of the square-root argument is enforced by replacing the argument with its maximum against zero, e.g., $\max(1 - \frac{4\epsilon}{q(x)}, 0)$. This produces the following **dynamic-adaptive clipping bounds for importance sampling**:

$$0.5 + \frac{1}{2}\sqrt{\max\left(1 - \frac{4\epsilon_{low}}{q(x)}, 0\right)} \leq r(x) \leq 0.5 + \frac{1}{2}\sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \quad (27)$$

For the upper and lower bounds in Inequality 27, we use separate hyperparameters ϵ_{high} and ϵ_{low} so that the adaptive interval aligns with the GRPO bound at selected reference points.

A.6 ANALYZING THE DIFFERENCES BETWEEN DYNAMIC-ADAPTIVE CLIPPING AND FIXED CLIPPING

To align the dynamic-adaptive interval with the fixed symmetric clipping interval at the boundary points $(\frac{1}{1+\epsilon}, 1)$ and $(1, 1 - \epsilon)$, as shown in Figure 4d, we solve the equations in Equation (28) for ϵ_{low} and ϵ_{high} .

$$\begin{cases} q(x) = \frac{1}{1+\epsilon} \\ 1 = 0.5 + \frac{1}{2} \sqrt{1 + \frac{4\epsilon_{high}}{q(x)}} \end{cases} \quad (28)$$

$$\begin{cases} q(x) = 1 \\ 1 - \epsilon = 0.5 + \frac{1}{2} \sqrt{\max(1 - \frac{4\epsilon_{low}}{q(x)}, 0)} \end{cases}$$

In the standard GRPO setting, the fixed clipping threshold is $\epsilon = 0.2$. Solving the calibration equations under this convention gives approximately $\epsilon_{low} = 0.16$ and $\epsilon_{high} = 0.2$. Although these values are unequal, their purpose differs from DAPO’s fixed asymmetric clipping: here the asymmetry calibrates a probability-dependent interval to match GRPO at specified points.

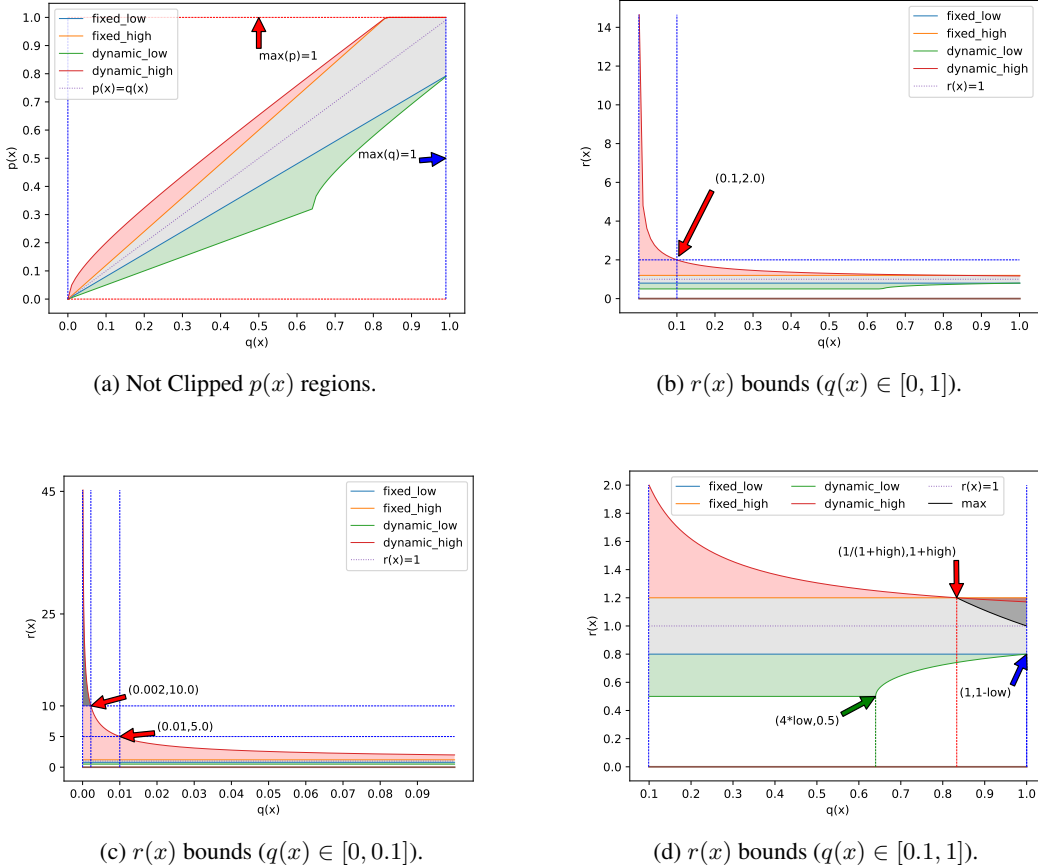


Figure 4: Comparison of fixed clipping ($\epsilon = 0.2$) and dynamic-adaptive clipping ($\epsilon_{low} = 0.16$, $\epsilon_{high} = 0.2$), showing how the adaptive bounds expand for low-probability tokens while matching the fixed interval at calibrated points.

Dynamic versus fixed clipping: Figure 4 contrasts the GRPO fixed clipping interval ($[1 - \epsilon, 1 + \epsilon]$) with the proposed probability-aware adaptive interval. For a token with old probability $q(x)$, fixed clipping assigns the same relative width 2ϵ regardless of probability; consequently, as $q(x)$ decreases, the absolute exploration space contracts linearly (Figure 4a). Dynamic adaptation instead makes the relative width inversely related to $q(x)$, which keeps the effective exploration space approximately stable across the probability range.

Effect of decreasing $q(x)$: When $q(x)$ becomes small, the fixed interval $[1 - \epsilon, 1 + \epsilon]$ remains unchanged and therefore does not increase the model’s ability to update low-probability tokens. The dynamic-adaptive bound expands rapidly in this regime (Figure 4b), creating a larger relative update space for rare tokens and directly supporting exploration-efficient RL.

Behavior for extremely low probabilities: Figure 4c focuses on the interval $q(x) \in [0, 0.1]$. As $q(x) \rightarrow 0$, the dynamic-adaptive upper bound for $r(x)$ grows rapidly according to the inverse square-root form, whereas the fixed-clipping upper bound remains constant. Following the dual-clipping strategy of Ye et al. (2020), we impose a hard ceiling $r_{\max} = 10$ to avoid instabilities such as gradient vanishing or explosion. The gray region in Figure 4c corresponds approximately to $q(x) \leq 2 \times 10^{-3}$, where the adaptive upper limit is capped at $r_{\max}$.

Upper and lower bounds: For the lower bound, when $q(x) \leq 4\varepsilon$, dynamic-adaptive clipping gives $dynamic_{low} = 0.5$, whereas fixed clipping retains the constant $fixed_{low} = 1 - \varepsilon = 0.8$. Once $q(x)$ exceeds this threshold, the dynamic-adaptive lower bound smoothly contracts toward the fixed value. The upper bound behaves analogously: it exceeds the fixed bound whenever $q(x) < \frac{1}{1+\varepsilon_{high}}$ and merges with it for larger $q(x)$. Thus, the adaptive interval $[dynamic_{low}(q(x), \varepsilon_{low}), dynamic_{high}(q(x), \varepsilon_{high})]$ widens the admissible region for rare tokens while preserving the tightness of fixed clipping for common tokens (Figure 4d).

Positive versus negative advantage: The loss in Equation (8) includes both positive- and negative-advantage cases. Under the minimum operation, a positive-advantage ratio $r(x)$ is clipped within $[0, dynamic_{high}(q(x), \varepsilon_{high})]$, whereas a negative-advantage ratio uses the interval $[dynamic_{low}(q(x), \varepsilon_{low}), \infty)$. Because excessively large $r(x)$ can destabilize training, we cap the ratio for both positive and negative advantages at $r_{\max} = 10$. This safeguard follows Ye et al. (2020) and prevents adaptive bounds from producing excessively high importance weights.

Overall, for both positive and negative advantages, dynamic-adaptive clipping gives low-probability rare tokens a larger admissible update region than fixed clipping, thereby increasing exploration capacity where it is most constrained.

A.7 THE DEFINITIONS OF THE ADVANTAGE STANDARDIZATION

Let i denote the iteration in which prompts are used to generate responses and update model parameters. The reward-normalization statistics used by SAS are defined as follows.

The response-reward means are given by the functions in Equation (29):

$$\begin{cases} \mu_{new}^i = \frac{1}{G} \sum_{j=1}^G R_j^i \\ \mu_{old}^i = \frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G R_j^k \\ \mu_{total}^i = \frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G R_j^k \\ \quad = \frac{1}{i} (\mu_{new}^i + (i-1) \mu_{old}^i) \end{cases} \quad (29)$$

The corresponding response-reward standard deviations are defined in Equation (30):

$$\begin{cases} \sigma_{new}^i = \sqrt{\frac{1}{G} \sum_{j=1}^G (R_j^i - \mu_{new}^i)^2} \\ \sigma_{old}^i = \sqrt{\frac{1}{G*(i-1)} \sum_{k=1}^{i-1} \sum_{j=1}^G (R_j^k - \mu_{old}^i)^2} \\ \sigma_{total}^i = \sqrt{\frac{1}{G*i} \sum_{k=1}^i \sum_{j=1}^G (R_j^k - \mu_{total}^i)^2} \\ \quad = \sqrt{\frac{1}{i} \left(\sigma_{new}^i{}^2 + (i-1) \sigma_{old}^i{}^2 + \frac{i-1}{i} (\mu_{old}^i - \mu_{new}^i)^2 \right)} \end{cases} \quad (30)$$

To reduce computational and memory cost when computing σ_{total}^i and μ_{total}^i in Equation (30) and Equation (29), we use equivalent expressions based on μ_{new}^i , σ_{new}^i , μ_{old}^i , and σ_{old}^i rather than recomputing the statistics directly from all original rewards.

A.8 THE TEMPLATE FOR EXPERIMENTS

All experiments use the following prompt format, adapted from the Qwen-Math template (Yang et al., 2024b), to keep training and evaluation formatting consistent.

Qwen-Math Template

```
<|im_start|>system
Please reason step by step, and put your final answer within \boxed{ }. <|im_end|>
<|im_start|>user
{problem}
<|im_end|>
<|im_start|>assistant
```

A.9 TRAINING SETTING

GRPO uses a symmetric clipping threshold $\epsilon = 0.2$. DAPO uses lower/upper clipping thresholds $(0.2, 0.28)$, an additional 512-token soft-punish cache, and a maximum probability ratio of 10 for negative advantages. GSPO uses lower/upper clipping thresholds $(3e - 4, 4e - 4)$. DCPO uses low/high clipping thresholds $(0.16, 0.2)$, preserving the GRPO clipping points $(q(x), r(x)) = (\frac{1}{1+\epsilon}, 1 + \epsilon)$ and $(q(x), r(x)) = (1, 1 - \epsilon)$ shown in Figure 4d, and sets the maximum probability ratio to 10 for both positive and negative advantages, unlike DAPO.

Training runs for 400 steps with response-generation batch size 512 and parameter-update mini-batch size 32. For each problem, we generate $G=16$ responses using temperature 1 and top_p 1. The prompt maximum length is 1,024 and the response maximum length is 3,072, substantially shorter than the 20k maximum length used in the original DAPO setting.

The training codebase is adapted from Verl (Sheng et al., 2024) and experiments are run on 32xH20 GPUs. Unless otherwise specified, hyperparameters follow DAPO. For loss computation, we retain the original $\frac{1}{G}$ or $\sum_{i=1}^G \frac{1}{|o_i|}$ weighted-average formulations from GRPO or DAPO rather than the approximate weighted calculation used in Verl, ensuring compatibility across different microbatch sizes.

A.10 THE DETAILED PERFORMANCE OF MODELS

Avg@1 Trend Figure 5 reports average Avg@1, computed by averaging greedy-decoding results over MATH500, AMC23, AIME24, and AIME25. DCPO consistently outperforms GRPO and DAPO across all four models and outperforms GSPO on the two models where GSPO is evaluated. Its trajectory is also more stable, avoiding the pronounced GRPO performance drop and the larger DAPO fluctuations. These trends indicate that DCPO improves deterministic reasoning performance while maintaining steadier optimization dynamics.

Avg@32 Trend Figure 6 compares Avg@32 trajectories for DCPO, GRPO, DAPO, and GSPO where available on AMC23, AIME24, and AIME25. GRPO shows substantial fluctuations and occasional collapse; on Qwen2.5-14B, its performance drops sharply after approximately 170 training steps. DCPO instead follows a consistently upward trajectory with limited fluctuation across all evaluated model sizes from 1.5B to 14B, including the early training phase. Relative to DAPO, DCPO attains stronger performance while using approximately half the training time and resources. These results support the combined role of dynamic-adaptive clipping and smooth advantage standardization in improving response-level exploration and policy-distribution robustness.

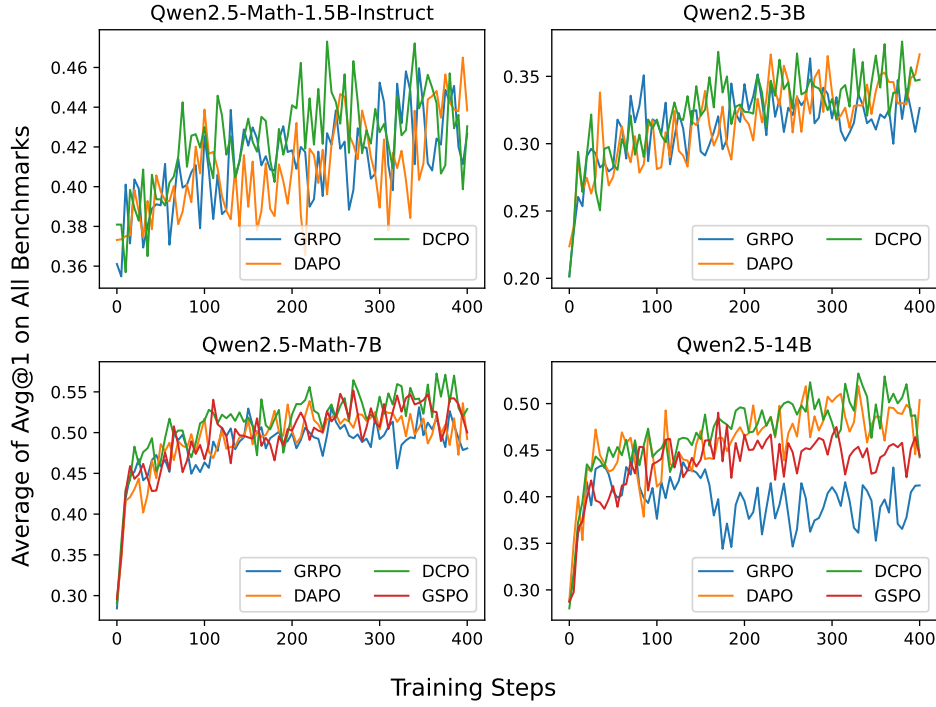


Figure 5: Average Avg@1 performance across MATH500, AMC23, AIME24, and AIME25 under greedy decoding.

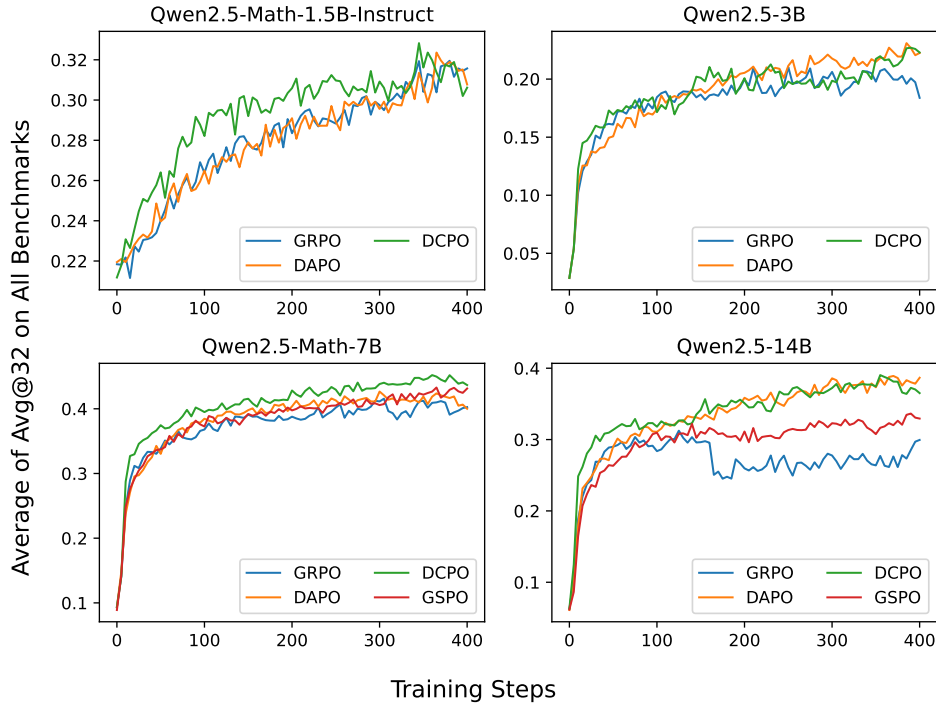


Figure 6: Average Avg@32 performance across AMC23, AIME24, and AIME25 under 32-response sampling.

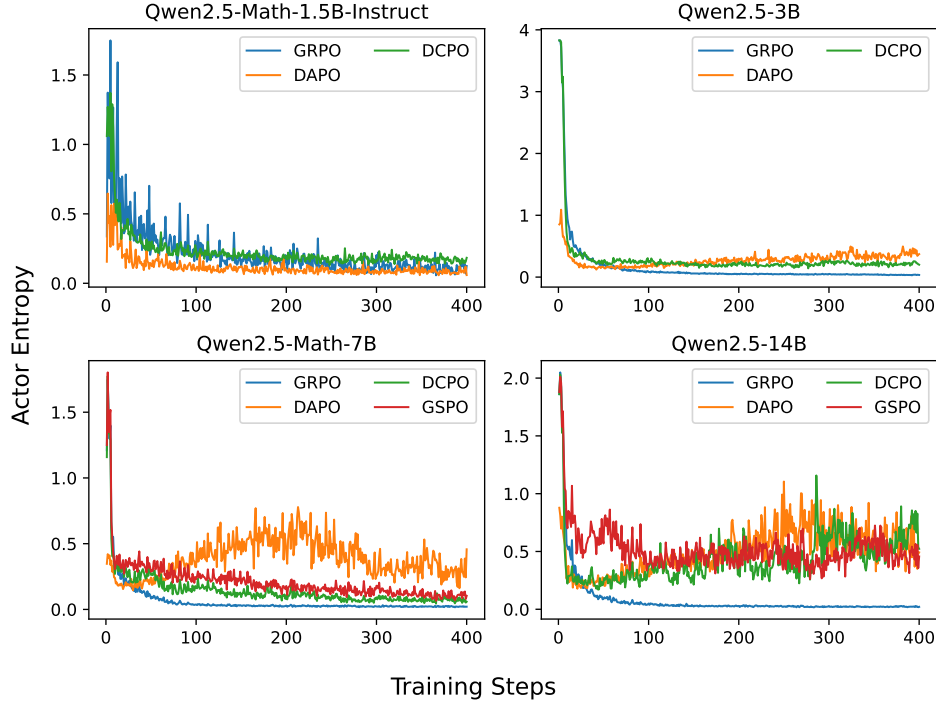


Figure 7: Model entropy during training, illustrating the exploration–stability regimes induced by each optimizer.

Entropy Trend Figure 7 shows that GRPO undergoes sharp entropy decay after an initial warm-up and then stabilizes in a flat, low-entropy regime. This entropy collapse reflects a rapid loss of policy randomness, reducing sampled-response diversity and limiting subsequent policy improvement.

DAPO and DCPO exhibit model-dependent entropy behavior. On the instruct model Qwen2.5-Math-1.5B, DAPO, DCPO, and GRPO follow broadly similar declining trajectories. On the larger base models, Qwen2.5-3B, Qwen2.5-Math-7B, and Qwen2.5-14B, the methods separate more clearly: DAPO maintains the highest entropy and shows an upward trend with large fluctuations. GSPO has entropy similar to DCPO on Qwen2.5-Math-7B and comparable average metric performance, but on Qwen2.5-14B its performance fluctuates downward. DCPO occupies an intermediate regime, with a smoother entropy curve bounded between GRPO’s low-entropy collapse region and DAPO’s high-variance region.

These observations suggest a U-shaped relationship between entropy and training performance. Excessively low entropy removes exploratory signal and encourages premature convergence, whereas excessively high or unstable entropy undermines training stability. DCPO maintains entropy in a moderate band, supporting both convergence and sufficient exploration; this balance becomes especially valuable as model capacity increases and is consistent with DCPO’s stronger inference performance across model sizes.